

MARC CASTRO

Tech Consultant | Lead AI Engineer | Forward Deployed Engineer

+63 919 902 5869 | MCastro.DSci@gmail.com

linkedin.com/in/nipmcastro | portfolio.marccastro.solutions

PROFESSIONAL SUMMARY

Lead AI Engineer and Forward Deployed Engineer who builds production LLM, RAG, and agentic AI systems that cut manual effort and processing time by up to 95%. 5+ years deploying AI across insurance, customer experience, education, and government. MS in Data Science; multiple-time national and international hackathon champion and published bioinformatics co-author.

PROFESSIONAL EXPERIENCE

Lead AI Engineer / Tech Consultant (Independent Contractor)

Sep 2024 – Present

Intelligent AI Solutions — UK-based AI Consultancy

- Delivered production LLM systems that replaced manual research workflows for enterprise clients, including an automated media-monitoring platform for bias detection and a RAG knowledge base for a global travel agency, accelerating analyst decision-making by at least 50%.
- Shipped agentic chatbots, no-code automations, and streamlined workflow systems across 10 industries by driving research into advanced RAG architectures and fine-tuning, cutting manual processing effort for clients.

Freelance Data Science / AI Automation Instructor & Mentor

Nov 2023 – Present

CIIT & Eskwelabs

- Upskilled over 200 professionals from government, corporate, and academic backgrounds by leading hands-on workshops, bootcamps, and hackathons on AI, No-Code AI, and Data Science.

Senior Data Science Consultant

Sep 2024 – Dec 2025

PSHS – Knowledge Innovation Division

- Enabled targeted academic interventions by building student-performance analytics that surfaced at-risk students earlier and informed data-driven curriculum enhancements.

Global Data Science Manager

Aug 2023 – Sep 2024

TDCX AI — Singapore

- Reduced agent time-to-proficiency by at least 20% by building an LLM-powered training app that simulated realistic customer conversations during onboarding.
- Streamlined workforce-management planning by delivering a forecasting model that unified predictive analytics and simulation in a single user-friendly interface.

Data Scientist

Aug 2022 – Aug 2023

AIA Digital+ — Technology arm of AIA, Asia-Pacific's largest listed life insurer

- Reduced Next Best Action model-development effort by up to 95% by designing an end-to-end automated data pipeline for AIA Indonesia campaigns.
- Cut projected claim-processing time by 25%+ by building an end-to-end fraud-detection solution that automated claim-request approvals.

Senior Research Associate (Analytics & Visualization)

Sep 2020 – Mar 2021

Philippine Genome Center

- Accelerated SARS-CoV-2 genomic reporting by developing custom data pipelines and analytical modules, cutting turnaround from 5 days to 3 hours.
- Co-authored three peer-reviewed papers on emerging SARS-CoV-2 variants in the Philippines.

EDUCATION

Master of Science in Data Science <i>Asian Institute of Management</i>	2022
Bachelor of Science in Applied Physics (Instrumentation) <i>University of the Philippines – Diliman</i>	2020

TECHNICAL SKILLS

Languages: Python, SQL
AI & ML: LLMs, RAG, fine-tuning, agentic workflows, predictive analytics, forecasting, fraud detection
Cloud & Data: Azure (AI Search, Container Apps, Azure AD, AI Foundry), Google Cloud (Vertex AI, Pub/Sub), CosmosDB, PostgreSQL, ChromaDB
Frameworks & Tools: FastAPI, AsyncIO, Pydantic, Cohere Reranker, Papermill, n8n

SELECTED PROJECTS

- Enterprise RAG Chat Platform — Reduced time-to-answer and support-ticket volume by giving knowledge workers instant, cited answers from internal libraries; built as a production RAG system on Azure Container Apps with semantic, vector, and hybrid retrieval, GPT-4V vision, role-based access control, and per-user credit limits. (Azure AI Search, Container Apps, Azure AD, CosmosDB, Azure AI Foundry, Python)
- Customer Service Agent Copilot — Lowered average handle time by surfacing real-time resolution paths to agents; a FastAPI copilot that analyzes live conversations and returns structured JSON scorecards from GPT-4o-mini. (Google Pub/Sub, GPT API, Cohere Reranker, Graph Database, AsyncIO, FastAPI)
- Document Intelligence Pipeline — Cut document processing from hours to minutes at 10–30x throughput via a four-phase async pipeline (Architect, Specialist, Builder, Verifier) converting complex multi-page PDFs to semantic HTML. (Vertex AI, ChromaDB, AsyncIO, Google Pub/Sub, Pydantic)
- Automated Media Analysis Pipeline — Eliminated daily manual reporting across multiple analyst teams by delivering fully automated, reproducible media analysis on a scheduled pipeline. (Papermill, Python, Vertex AI, PostgreSQL, Model Drift Monitoring)

AWARDS & RECOGNITION

- ICSC Hack-A-Street 2025 — Champion (National)
- DOH-DAP Hack4Health 2024 — Champion (National)
- UN Big Data Hackathon 2023, Youth Track — 2nd Place of 122 teams (International); Best Youth Team in Asia
- Usher Hackathon 2022 — 3rd Place (National)

PUBLICATIONS

- Detection and genome sequencing of SARS-CoV-2 belonging to the B.1.1.7 lineage in the Philippines. doi:10.1128/MRA.00219-21
- Genome sequencing and analysis of an emergent SARS-CoV-2 variant from the Central Visayas Region of the Philippines. doi:10.1101/2021.03.03.21252812
- A directed minimum connected dominating set for protein-protein interaction networks. ISBN:9780367814656

SELECTED SPEAKING & PUBLIC ENGAGEMENTS

- AI Technology Government Trainings — Department of Information and Communication Technology (2025)
- Generative AI & Low-Code for Corporate Teams — Ayala Group (2025)

- n8n Automation Tools for E-Commerce — Carousell Singapore (2025)
- Breaking into Tech Consulting within 5 Years — Asian Institute of Management (2025)
- BANT.AI: Revolutionizing Fraud Detection with Unified Health Data — Department of Health (2024)
- The Art of Data Storytelling and Data Visualization — University of Santo Tomas (2023)